

Professional Project Overview for Instructors

Universal Meaning-Based Language (UMBL)

This document provides an academic overview of UMBL (Universal Meaning-Based Language) for instructors, researchers, and course supervisors.

UMBL is an early-stage **language-design development project**, not a completed language.

Its long-term ambition is to explore whether a **pictographic, structurally organized, and evolvable constructed language**—analogous in spirit to Esperanto but based on pictographic principles—can be systematically designed.

UMBL currently functions as a **language engineering prototype**, intended for research, experimentation, and critical evaluation rather than for practical communication.

1. Background and Research Motivation

The conceptual foundation of UMBL originates from comparative observations of Chinese and English, especially in how both languages transition from everyday usage to academic expression:

- Chinese relies on character formation, semantic radicals, and visual structure.
- English relies on morphological construction, roots, and affixes.
- Chinese media culture often incorporates subtitles and visual overlays (e.g., “danmu”), reflecting a strong visual-semantic tendency.
- Chinese has a historical tradition of “character creation,” while English tends toward “root creation.”
- Pictographic or semi-pictographic expression appears more natural in certain Chinese contexts.
- Phonetic writing systems face challenges in cross-cultural comprehension.
- Natural languages exhibit inconsistencies in abstraction, logic, and semantic transparency.

These observations motivate the central research question of UMBL:

Is it possible to design a pictographic, structurally transparent, and evolvable constructed language that offers new possibilities for clarity, learnability, and cross-cultural comprehension?

UMBL does **not** claim superiority over existing languages.

Its purpose is to explore an alternative design direction in language engineering.

2. Project Components

2.1 Pictographic Component Repository

(semantic_symbol_parts_warehouse)

- A modular collection of pictographic components (“semantic building blocks”).
- Each component encodes a specific semantic intention.
- Components can be combined to form more complex pictographic expressions.
- Serves as the foundation for future word- and sentence-level structures.

2.2 Language Structure Development (Exploratory Stage)

UMBL is actively investigating:

- How to define **pictographic morphemes**
- How to construct **pictographic words** through component combination
- How to design **pictographic sentence structures**
- How to formalize **grammar principles** suitable for a pictographic language
- How to define **phonology, evolution rules, and semantic extension mechanisms**
- How to maintain consistency and expandability over time

These ideas are discussed in the UMBL whitepaper but remain open to revision.

2.3 Whitepaper and Design Guidelines

The whitepaper outlines:

- Limitations of phonetic writing systems
- Structural inconsistencies in natural languages
- UMBL's design principles
- Proposed rules for character formation, pronunciation, grammar, and language evolution
- The rationale for engineering-oriented language construction

These documents provide theoretical grounding for academic use.

3. Academic Relevance (Cautiously Stated)

UMBL's academic value lies in its role as a **research object**, not as a finished language.

It intersects with multiple disciplines:

3.1 Linguistics

- Constructed language design methodologies
- Pictographic writing systems and semantic transparency
- Language evolution and engineered language frameworks

3.2 Cognitive Science

- Recognition and comprehension of pictographic symbols
- Learning curves for visual-semantic structures
- Misinterpretation patterns and cognitive load

3.3 Information Design & Standardization

- Modular symbol design
- Structural comparison with ISO graphical standards
- Consistency and readability in visual communication

3.4 Computer Science

- Encoding schemes for pictographic languages
- Automated symbol composition
- Machine-readable pictographic structures

UMBL provides a platform for experimentation rather than definitive conclusions.

4. Applications in Teaching and Student Research

UMBL is suitable for:

4.1 Undergraduate Thesis Topics

- Structural analysis of pictographic language prototypes
- Small-scale user studies
- Cross-system comparisons
- Prototype construction (symbol sets, grammar experiments)

4.2 Coursework and Classroom Discussion

- Case studies in language design
- Exercises in symbol construction
- Interdisciplinary research examples

4.3 Research Method Training

- Literature review
 - Experimental design
 - Data analysis
 - Language modeling
-

5. Project Limitations (Important)

To maintain academic rigor, UMBL explicitly acknowledges its limitations:

- It is **not** a mature language.
- Grammar and phonology are still exploratory.
- No large-scale user testing has been conducted.
- It is a **language prototype**, not a finalized system.
- It is suitable for research, not for practical communication.

These limitations also create valuable research opportunities.

6. Conclusion

UMBL is an open, transparent, and experimental **pictographic language-design development project**. Its significance lies not in presenting a finished language, but in:

- Providing students with analyzable, testable, and research-ready materials

- Offering instructors a cross-disciplinary teaching example
- Creating a platform for exploring the structure and cognition of pictographic language

UMBL's progress depends on ongoing academic discussion, critique, and experimentation.

Researchers and instructors are invited to engage with the project in a critical and exploratory manner.